

Team Worksheet — Lab 10

Sampling Strategy, Sample Size and Rational Subgrouping

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-10-sampling-frame.xlsx`.

3. What we produced

Deliverable: Calculated sample sizes for your capstone metrics and a rational subgrouping scheme.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap the four sampling schemes in full: simple random, stratified, systematic and cluster — with the failure mode of each.

Step 2. Identify the sampling bias risks in your capstone: convenience sampling, day-shift-only sampling, and sampling only when the process is running well.

Step 3. Recap the continuous-data sample size formula $n = (1.96 s / d)^2$, where s is the estimated standard deviation and d is the precision you require.

Step 4. Apply it to your project Y. Estimate s from historical data or a pilot sample, choose your required precision d , and calculate n .

Step 5. Recap the discrete-data formula $n = (1.96/d)^2 \times p(1-p)$, where p is the estimated proportion defective.

Step 6. Apply it to your attribute metrics. Note how much larger n becomes — this is the cost of measuring in pass/fail.

Step 7. Now design rational subgroups. A subgroup must contain only common-cause variation, so that between-subgroup variation shows up as a signal.

Step 8. Decide what varies WITHIN your subgroup and what varies BETWEEN subgroups. Getting this backwards makes control limits so wide that nothing ever signals.

Step 9. Set subgroup size and sampling frequency for your capstone Y, and state what shift size you would be able to detect.

Step 10. Document the sampling plan and check it against the data collection plan from Lab 9 for consistency.

7. Self-check

Your sample sizes are calculated from a real estimate of s or p , your subgrouping scheme places common-cause variation within subgroups, and you can state the shift size your plan can detect.

Complete? ■ Yes ■ Not yet

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